Technical Report

Project #4: Interpret
FEMA National Risk Index Sensitivity Analysis

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1. Introduction

Risk Averse, LLC, an independent risk analysis consulting research group, retained our engineering team to perform a sensitivity analysis of the Federal Emergency Management Agency (FEMA) National Risk Index (NRI) scoring methodology. The NRI is a nationwide dataset developed by the Department of Homeland Security that quantifies the relative risk of natural disaster events for every U.S. Census Tract and county using three primary components: Expected Annual Loss (EAL) from 18 natural hazard types, Social Vulnerability (SoVI), and Community Resilience (RESL). Communities are assigned qualitative ratings from Very Low to Very High based on their national percentile ranking (FEMA, 2025).

The client has asked our team to develop an independent alternative risk definition, apply it to two states, and determine whether categorical bias exists in how the NRI classifies communities. This report presents our methodology, results, and recommendations based on a complete analysis of California (9,106 Census Tracts across 58 counties) and Colorado (1,447 Census Tracts across 64 counties). These states were selected because they represent contrasting risk profiles: California faces diverse multi-hazard exposure including earthquakes, wildfires, coastal flooding, and drought, while Colorado faces a narrower hazard portfolio dominated by hail, winter weather, and wildfire.

2. Data Sources

Three primary datasets were used. The NRI Census Tract datasets were downloaded from the FEMA National Risk Index Data Resources page (https://hazards.fema.gov/nri/data-resources) for California (9,106 tracts, 467 columns) and Colorado (1,447 tracts, 467 columns). The ATSDR Social Vulnerability Index (SVI) datasets were downloaded from the CDC/ATSDR SVI page (https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html) for California (9,109 tracts, 158 columns) and Colorado (1,447 tracts, 158 columns). NRI County-level data was also downloaded for both states (California: 58 counties; Colorado: 64 counties) for geographic scale comparison.

3. Methods

3.1 NRI Risk Calculation

The NRI calculates risk using a multiplicative formula: Risk Index Score = Expected Annual Loss Score × Social Vulnerability Score ÷ Community Resilience Score. Each component is independently scored and converted to a national percentile. Qualitative ratings are assigned using thresholds: Very Low (≤20th percentile), Relatively Low (20th–40th), Relatively Moderate (40th–60th), Relatively High (60th–80th), and Very High (>80th percentile) (FEMA, 2025). Because the formula divides by Community Resilience, communities with high resilience scores receive lower risk ratings even when expected annual loss is substantial.

3.2 Alternative Risk Definition

Our alternative risk score uses a weighted additive model: Alternative Risk Score = 0.6 × EAL_SPCTL + 0.4 × (RPL_THEMES × 100). EAL_SPCTL is the NRI Expected Annual Loss state percentile (0–100) and RPL_THEMES is the ATSDR SVI overall percentile (0–1), multiplied by 100 to match. The 0.6 weight reflects the client's emphasis on hazard exposure as the primary driver of risk, while 0.4 captures social vulnerability using an independent CDC index. This approach replaces the multiplicative formula with an additive combination and substitutes the NRI's internal SoVI with the CDC/ATSDR SVI.

3.3 Data Processing

The NRI and SVI datasets were merged using the 11-digit Census Tract FIPS code. All NRI tracts matched SVI records. Sentinel values of -999 were replaced with NaN. After cleaning, 65 California tracts and 16 Colorado tracts were excluded, leaving 9,041 valid California tracts and 1,431 valid Colorado tracts. All analysis was performed in Python 3 using pandas, NumPy, matplotlib, seaborn, and GeoPandas.

4. Results and Discussion

4.1 Rating Distribution Comparison

Figure 1 compares the distribution of risk ratings between the NRI and alternative methods. In California, the NRI centers on Relatively Moderate (3,406 tracts) and Relatively High (2,630 tracts). In Colorado, the NRI concentrates 75.4% of tracts in Very Low and Relatively Low. The alternative method distributes tracts uniformly within each state by design. This reveals that the NRI's national ranking compresses Colorado's risk distribution relative to California.

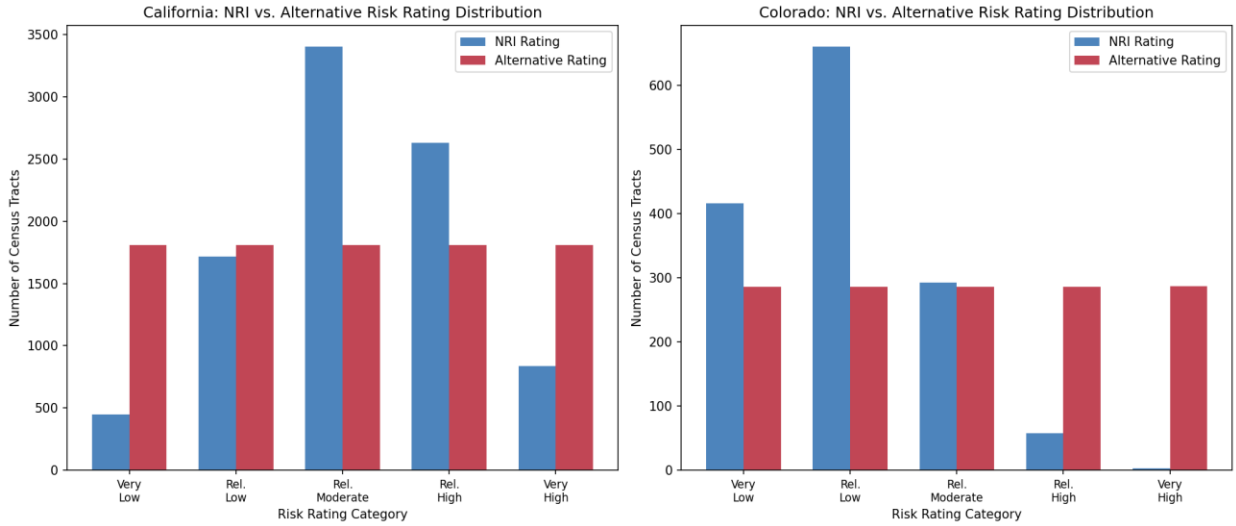

Figure 1. NRI vs. Alternative Risk Rating Distribution — California (left) and Colorado (right)

4.2 Cross-Tabulation Analysis

Tables 1 and 2 present the cross-tabulation of NRI ratings against alternative ratings. Diagonal cells (green) show agreement; off-diagonal cells show disagreement. Cells above the diagonal indicate the alternative method rates higher.

Table 1. California — NRI vs. Alternative Rating Cross-Tabulation (n = 9,041)

NRI \ Alt	V. Low	R. Low	R. Mod	R. High	V. High	Total
Very Low	**427**	23	0	0	0	450
Rel. Low	1157	**527**	35	0	0	1719
Rel. Moderate	224	1257	**1807**	117	1	3406
Rel. High	0	1	153	**1395**	1081	2630
Very High	0	0	0	109	**727**	836
Total	**1808**	**1808**	**1995**	**1621**	**1809**	**9041**

California: **4,336 of 9,041 tracts (48.0%)** agree. 3,081 (34.1%) downgraded, 1,624 (18.0%) upgraded by the alternative method.

Table 2. Colorado — NRI vs. Alternative Rating Cross-Tabulation (n = 1,431)

NRI \ Alt	V. Low	R. Low	R. Mod	R. High	V. High	Total
Very Low	**276**	133	7	0	0	416
Rel. Low	10	**153**	270	189	39	661
Rel. Moderate	0	0	**9**	89	195	293
Rel. High	0	0	0	**8**	50	58
Very High	0	0	0	0	**3**	3
Total	**286**	**286**	**286**	**286**	**287**	**1431**

Colorado: **445 of 1,431 tracts (31.1%)** agree. 976 (68.2%) upgraded, only 10 (0.7%) downgraded. This asymmetry is the strongest evidence of categorical bias.

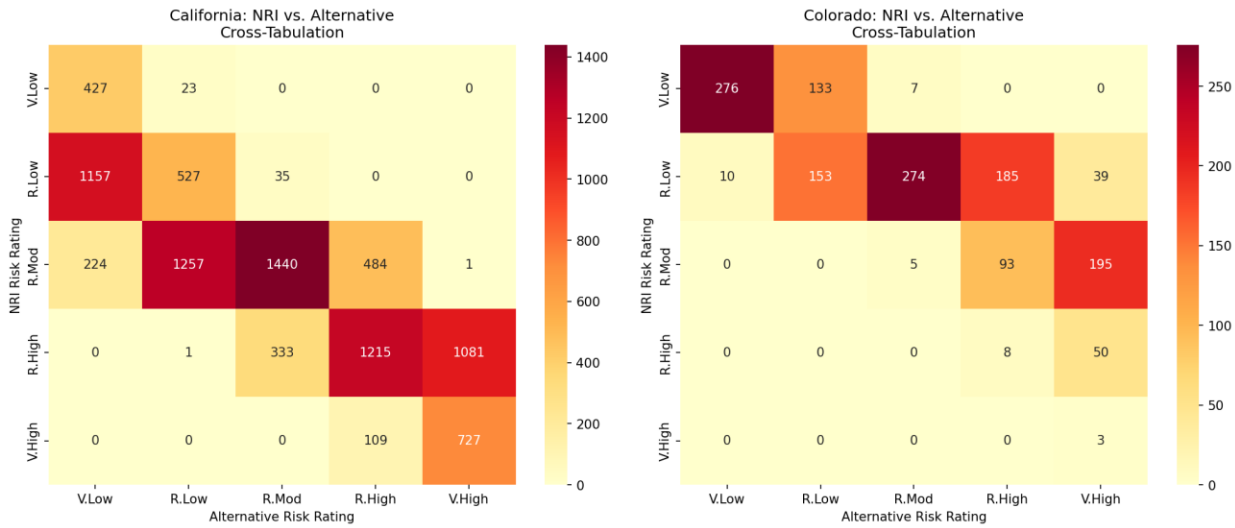

Figure 2. Cross-Tabulation Heatmap — California (left) and Colorado (right). Colorado shows values shifted above the diagonal.

4.3 Categorical Bias Direction

Figure 4 summarizes the direction of disagreement. In California, 48.0% agree, 18.0% are upgraded, and 34.1% are downgraded. In Colorado, only 31.1% agree while 68.2% are upgraded and less than 1% downgraded. The alternative method consistently assigns higher risk to Colorado communities.

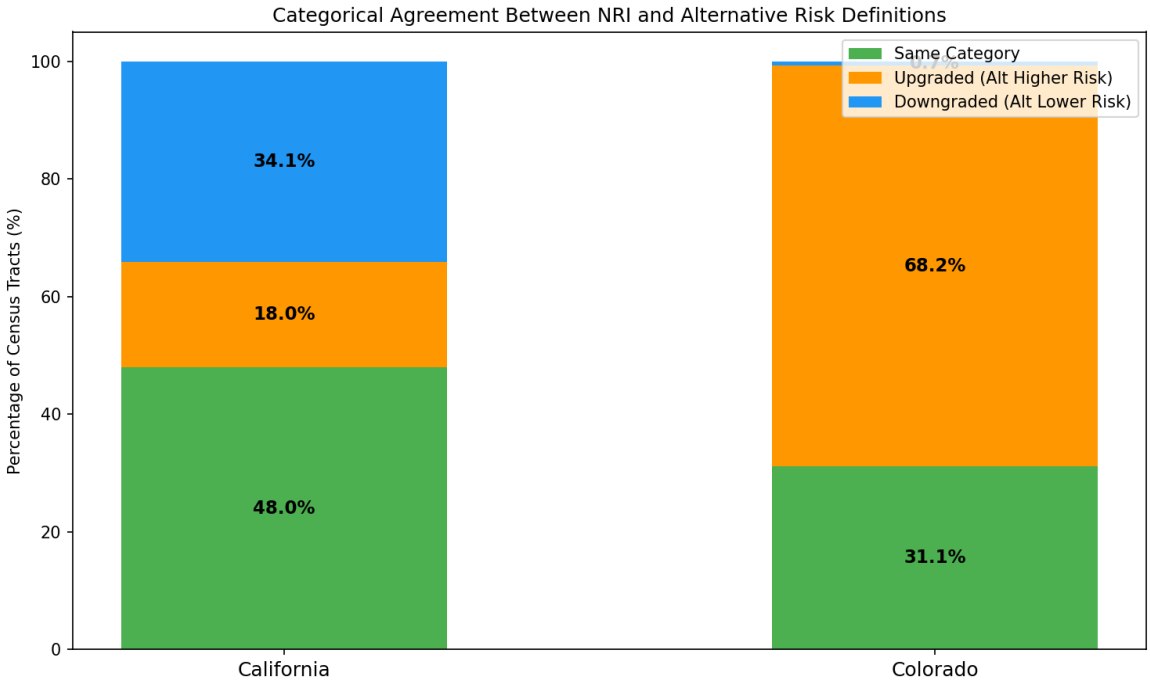

Figure 3. Categorical Agreement Between NRI and Alternative Risk Definitions

4.4 Geographic Scale Bias

Figure 5 compares NRI ratings between county and Census Tract levels. In Colorado, 71.9% of counties are rated Very Low versus 29.6% of tracts. County aggregation further compresses risk and obscures vulnerable communities. This confirms that geographic scale itself introduces categorical bias.

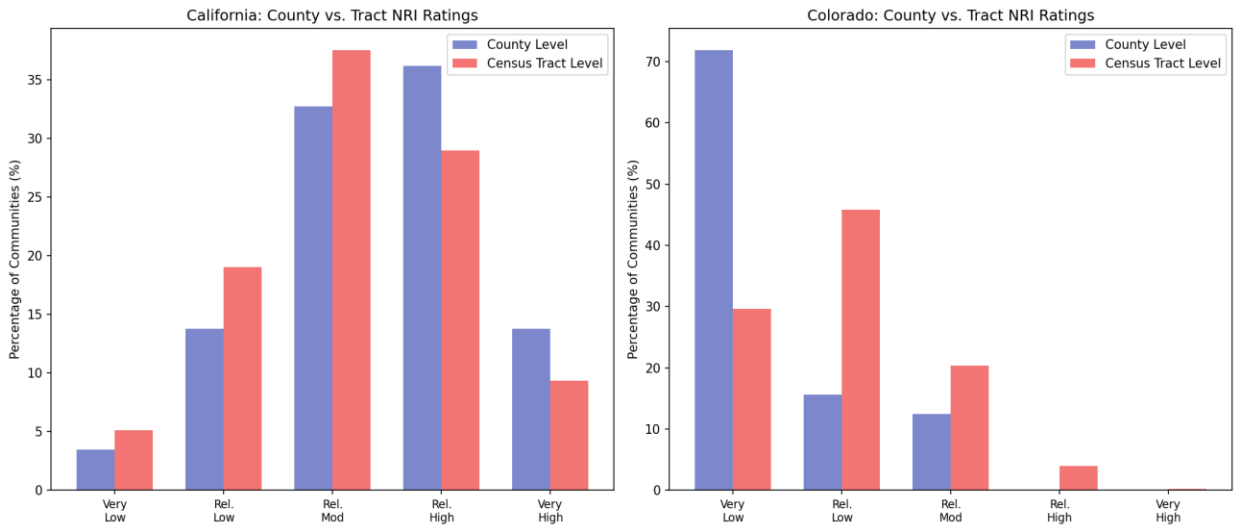

Figure 4. County-Level vs. Census Tract NRI Ratings — California (left) and Colorado (right)

4.5 EAL vs. SVI Component Analysis

Figure 3 plots Expected Annual Loss percentile against SVI percentile, colored by NRI rating. California shows a clear gradient; Colorado shows many tracts with high SVI but low NRI ratings, suggesting the multiplicative formula suppresses risk for socially vulnerable communities with lower expected loss.

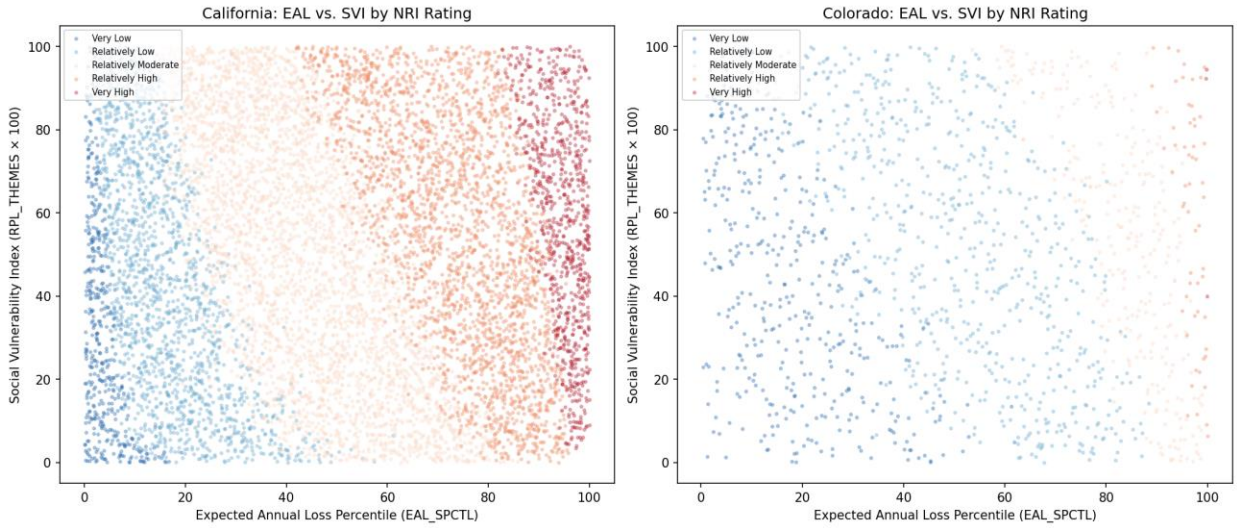

Figure 5. EAL vs. SVI by NRI Rating — California (left) and Colorado (right)

4.6 SVI Theme Analysis of Disagreement Tracts

Figure 6 compares mean SVI theme percentiles between agreement and disagreement tracts. In both states, disagreement tracts have higher SVI values across all four themes, with the largest gaps in Socioeconomic Status and Housing/Transportation. The communities most affected by categorical bias are those with the highest social vulnerability.

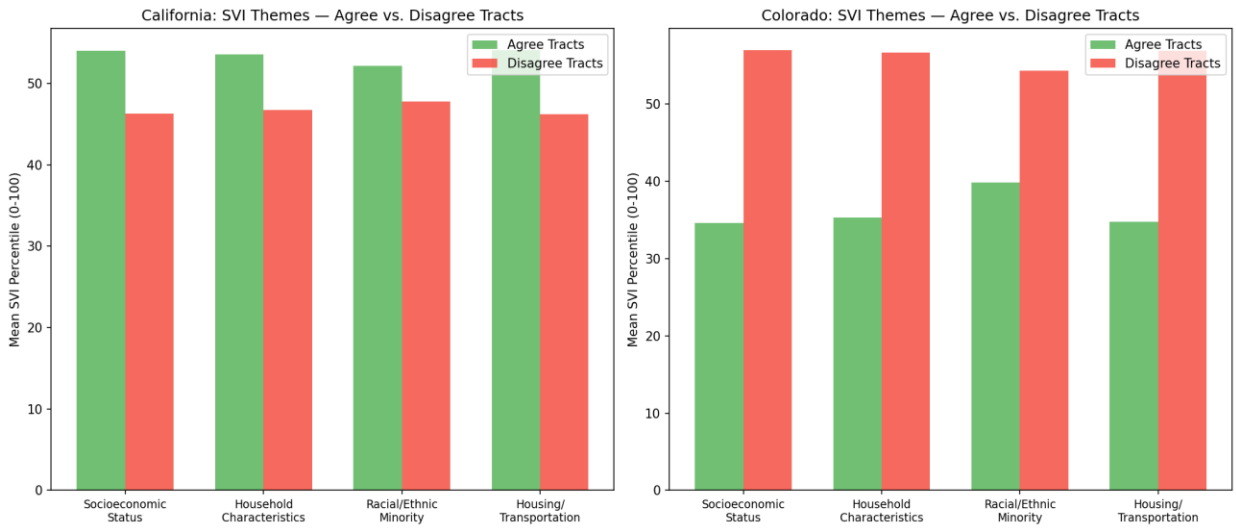

Figure 6. Mean SVI Theme Percentiles — Agreement vs. Disagreement Tracts

4.7 EAL vs. Community Resilience

Figure 7 shows the relationship between EAL and Community Resilience, colored by alternative risk score. In Colorado, nearly all tracts cluster at low EAL values, but the color gradient reveals that social vulnerability creates meaningful risk differentiation that the NRI's resilience denominator suppresses.

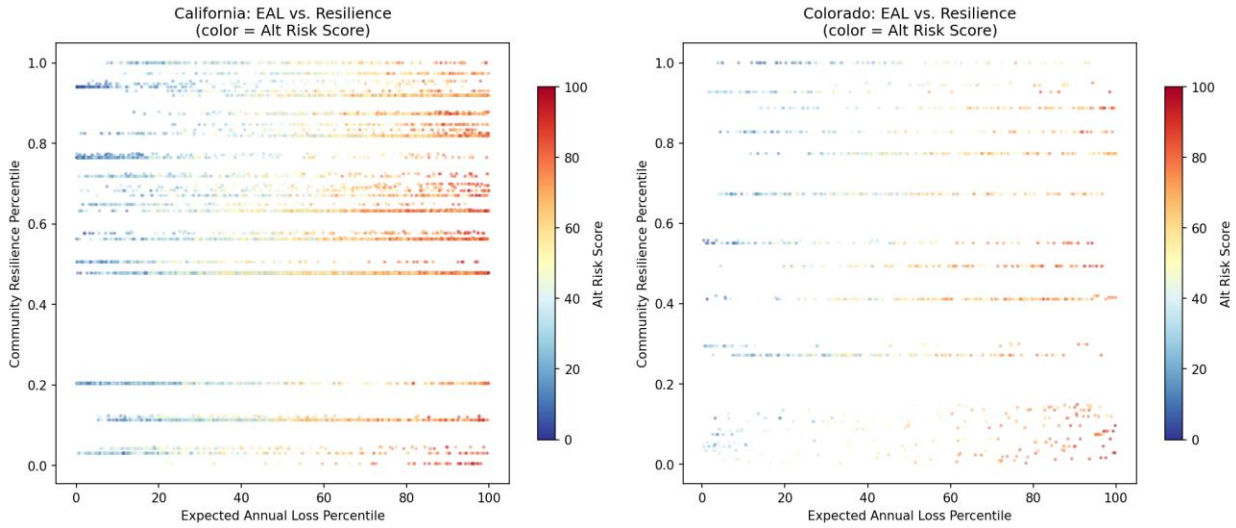

Figure 7. EAL vs. Community Resilience, colored by Alternative Risk Score

5. Ethical Implications and Bias Analysis

The NRI's multiplicative formula and national ranking system implicitly prioritize communities facing the largest absolute expected losses in high-population, high-property-value areas. Communities in states like Colorado, where property values and population densities are lower but social vulnerability is high, receive systematically lower risk ratings despite genuine hazard threats.

Figure 6 demonstrates that the communities most affected by categorical bias have the highest social vulnerability scores across all four CDC/ATSDR themes. The NRI may inadvertently penalize the very communities that disaster mitigation programs are designed to help. A rural Colorado community experiencing severe hail damage may be classified as Relatively Low risk due to low property values, even though the human impact relative to that community's limited resources could be devastating.

The use of the CDC/ATSDR SVI as an independent vulnerability measure captures poverty, lack of vehicle access, crowded housing, and minority status—factors that directly affect evacuation, sheltering, and recovery. Neither definition is objectively correct; each encodes different assumptions about whose risk matters most. Transparency about these assumptions is essential for equitable disaster mitigation policy.

6. Implications for FEMA Funding

In Colorado, the NRI rates only 61 tracts (4.3%) as Relatively High or Very High, while the alternative method rates 573 tracts (40.0%) in these categories. Under the NRI, the vast majority of Colorado communities may be excluded from FEMA mitigation funding despite meaningful hazard exposure and high social vulnerability. In California, the two methods show much closer agreement (38.5% vs. 40.0% in the top two categories).

This disparity suggests NRI-based funding decisions may concentrate resources in high-absolute-loss states at the expense of states where social vulnerability is the primary risk driver. A community that never receives mitigation funding because its NRI rating is Relatively Low will be less prepared and slower to recover when disaster strikes.

7. Conclusion and Recommendations

This sensitivity analysis demonstrates that categorical bias exists in the FEMA NRI. California shows 48.0% agreement between methods; Colorado shows only 31.1%, with the alternative method upgrading 68.2% of tracts. We recommend: (1) Test any risk definition for cross-state bias before using it for funding. (2) Consider the CDC/ATSDR SVI as a supplement to the NRI's internal vulnerability measure. (3) Make the choice between multiplicative and additive formulas transparently with community input. (4) Do not use county-level data as a substitute for tract-level analysis.

8. References

ATSDR. (2022). CDC/ATSDR Social Vulnerability Index (SVI). Agency for Toxic Substances and Disease Registry. https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html

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